CS2201 Computer Organization & Architecture

L21: Write Strategies in Cache Memory



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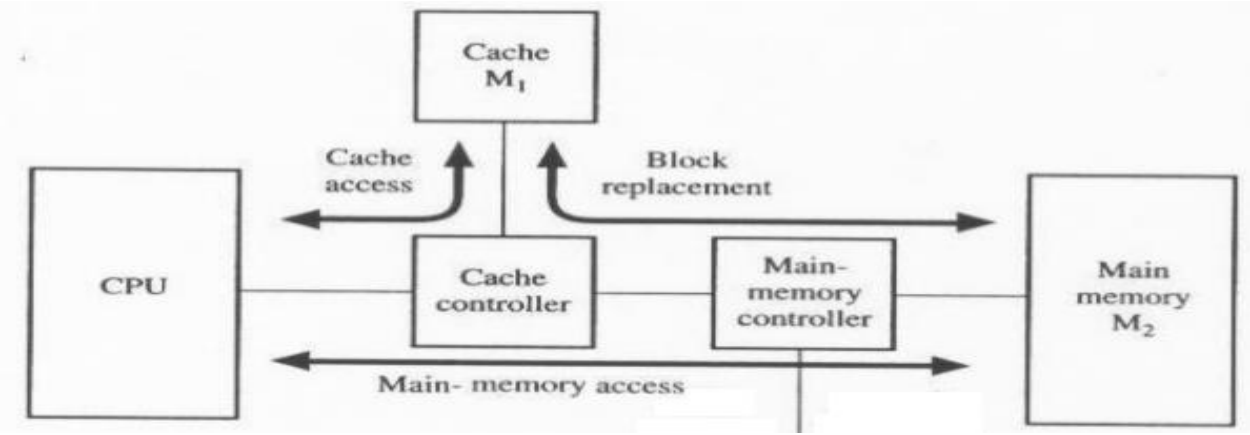
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Four cache memory design choices

- Where can a block be placed in the cache?
  - Block Placement
- How is a block found if it is in cache memory?
  - Block Identification
- Which block should be replaced on a miss?
  - Block Replacement
- What happens on a write?
  - Write Strategy

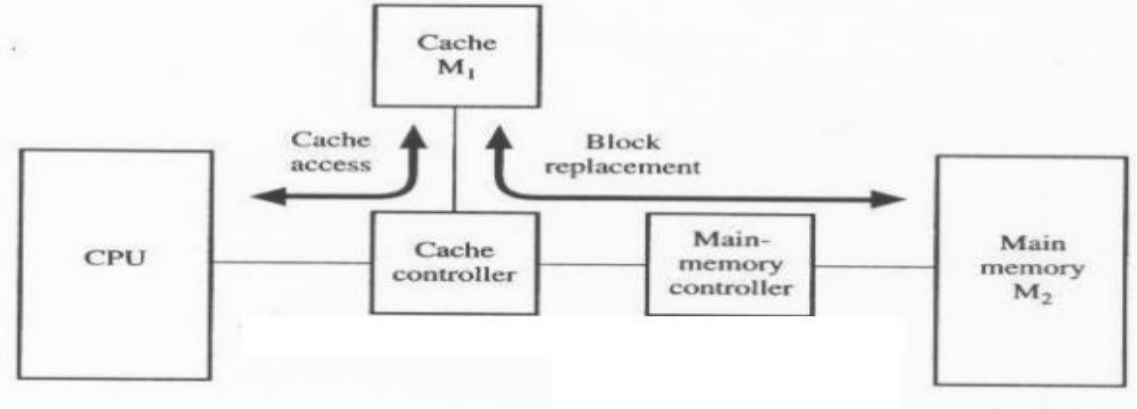
Look-aside vs Look through caches

- **Look-aside cache:** Request from processor goes to cache and main memory in parallel
- Cache and main memory both see the bus cycle
- On cache hit→ processor loaded from cache, bus cycle terminates; On cache miss: processor & cache loaded from memory in parallel



Look-aside vs Look through caches

- **Look-through cache:** Cache checked first when processor requests data from memory
- On hit→ data loaded from cache
- On miss→ cache loaded from memory, then processor loaded from cache

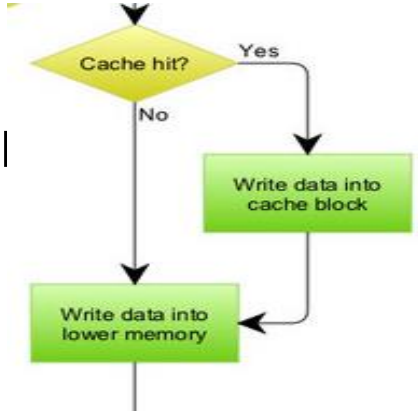


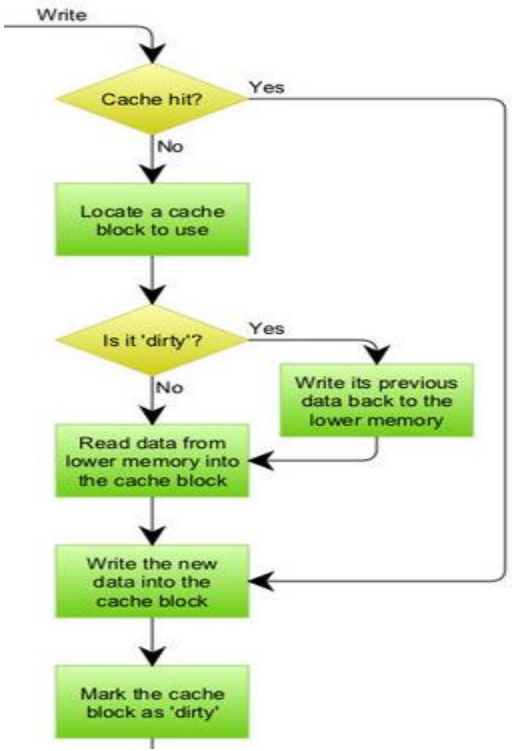
Write strategy

- **Write Hits → Write through vs Write back**
- **Write Miss→ Write allocate vs No-Write allocate**
- **Write through:** The information is written to both the block in the cache and to the main memory
- Read misses do not need to write back evicted line contents
- **Write back:** The information is written only to the block in the cache. The modified cache block is written to main memory only when it is replaced.
- Have to maintain whether block clean or dirty. No extra work on repeated writes; only the latest value on eviction gets updated in main memory.

Write strategy

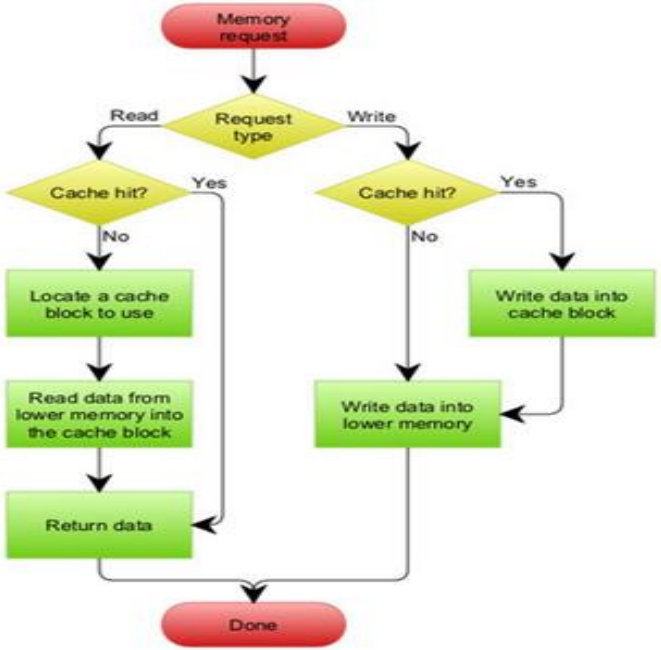
- **Write allocate:** The block is loaded into cache on a write miss.
- Used along with write back caches
- **No-Write allocate:** The block is modified in the main memory but not in cache
- Used along with write th
caches



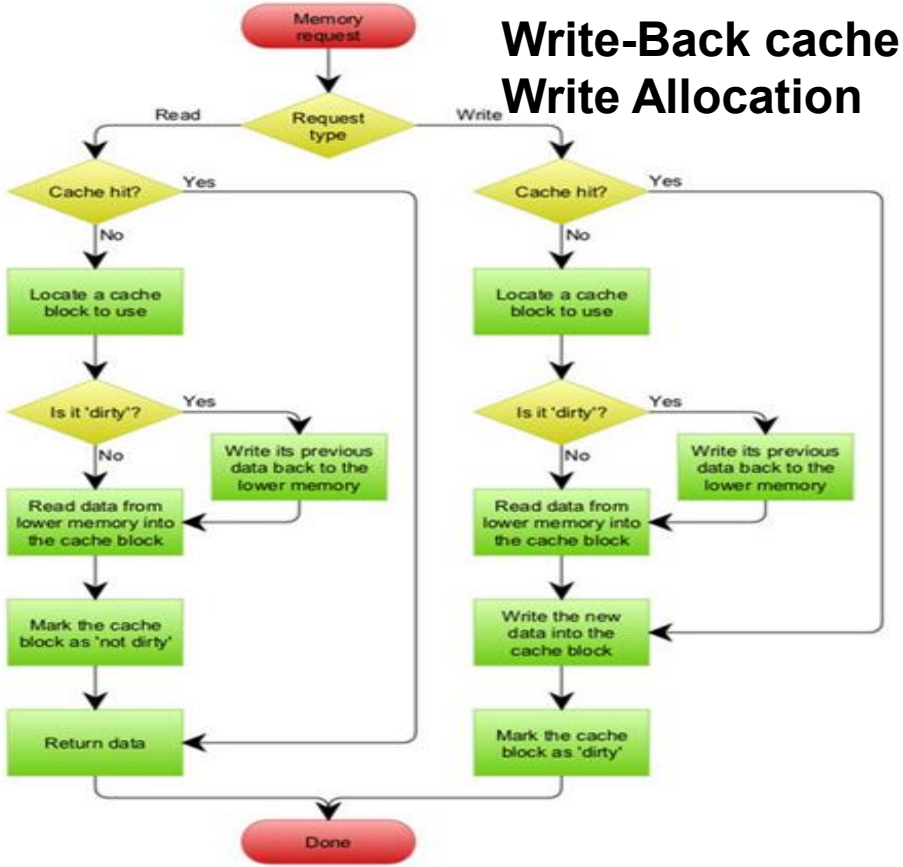


Write strategy

Write-Through cache with No-Write Allocation



Write-Back cache with Write Allocation



Types of Cache Misses

- **Compulsory**
  - Very first access to a block
  - Will occur even in an infinite cache
- **Capacity**
  - If cache cannot contain all the blocks needed
  - Misses in fully associative cache (due to the capacity)
- **Conflict**
  - If too many blocks map to the same set
  - Occurs in associative or direct mapped cache

Cache Block Concepts

For a 32KB direct mapped cache with 64 byte cache block, give the address of the starting byte of the first word in the block that contains the address 0X 7245E824.

Cache Block Concepts

32KB direct mapped cache with 64 byte cache block address 0X 7245E824.

sets = CS/(BSxA) = $2^{15}/(2^6 x 1) = 2^9$ = 512 sets

Tag	Index =9	Offset=6

Address: 0x 7245E824

----------0010 0100 offset (6 bits)

Block address range from ----- 0000 0000 to ----- 0011 1111

0x 7245E800 to 0x 7245E83F

Hence starting address is 0x 7245E800

Index and Offset Calculations

A cache has 512 KB capacity, 4B word, 64B block size and 8-way set associative. The system is using 32 bit address. Given the following addresses, which set of cache will be searched and specify which word of the selected cache block will be forwarded if it is a hit in cache?
(a) 0X ABC89984 (b) 0X 485669AC

Index and Offset Calculations

A cache has 512 KB capacity, 4B word, 64B block size and 8-way set associative. The system is using 32 bit address.
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sets = CS/(BSxA) = $2^{19}/(2^6 x 2^3) = 2^{10}$ = 1024 sets
1 word = 4B , Hence 64 byte block has 16 words

Tag = 16	Index =10	Offset=6 (4+2)

0x ABC89984 = 1001 1001 1000 0100 □ Set 614, word 1

0x 485669AC = 0110 1001 1010 1100 □ Set 422, word 11

Tag and Data array Access

It takes 3 ns to access a tag array value and 4.2 ns to access a data array, 1.3 ns to perform hit/ miss comparison and 1ns to return the selected data to processor.

(a) What is the cache hit latency of the system?

(b) What is the cache hit latency of the system if hit/ miss comparison will take 0.6 ns?

(c) What would be the hit latency if both tag and data array access take 3.5 ns and hit/ miss comparison take 1 ns?

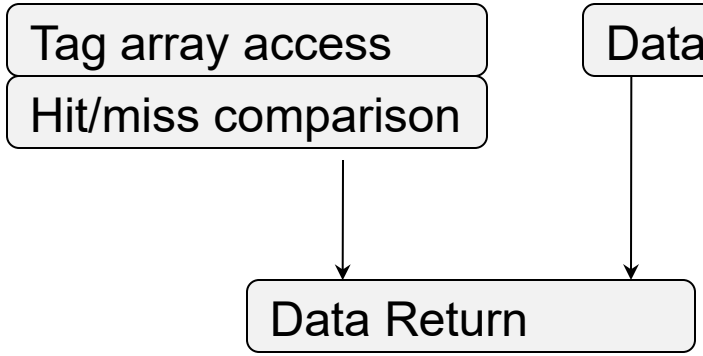
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(a) Max (3+1.3, 4.2) + 1 =5.3ns

(b) Max (3+0.6, 4.2) + 1 =5.2ns

(c) Max (3.5+1, 3.5) + 1 =5.5ns



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